Code No. 8411

OSMANIA UNIVERSITY
FACULTY OF ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (AUTONOMOUS)
B.E. (CSE, AI&ML) IV-Semester (MAIN) Examinations July-2025

1005237290O8

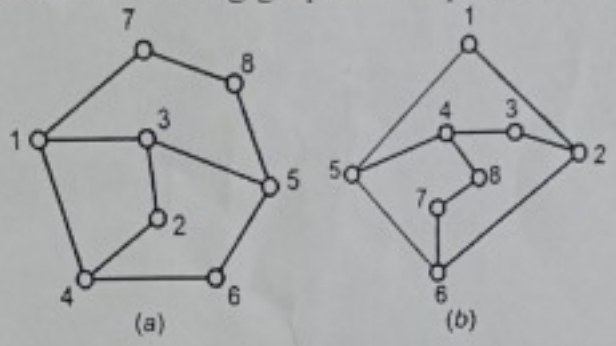
GRAPH THEORY

Time : 3 hours **Max. Marks : 60**

Note : *i) Answer **Question No. 1** (Compulsory) and answer any **four questions** from the remaining questions (2- 7). Each question caries 12 marks.*
ii) Answers must be written in same order as they occur in the Question Paper.
*iii) **Missing data, if any may suitably be assumed.***

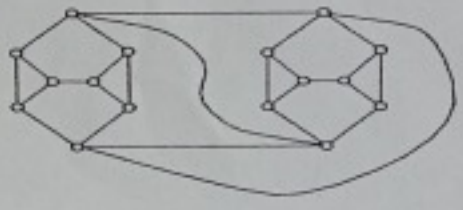
		Marks	BT	CO
1. a)	How many labeled trees are there with n vertices? Draw all labeled trees with 3 vertices.	2	2	1
b)	Show that all vertices of an Euler graph G are of even degree.	2	2	2
c)	Define covering.	2	1	1
d)	Prove that the chromatic polynomial of a complete graph with 4 vertices is $\lambda(\lambda - 1)(\lambda - 2)(\lambda - 3)$	2	2	2
e)	State the importance of Euler's formula.	2	1	1
f)	There are 25 telephones in Metropolis. Is it possible to connect them with wires so that each telephone is connected with exactly 7 others? Why?	2	3	3
2. a)	Write Depth first Search algorithm for minimal spanning tree and trace with a suitable example.	6	2	3
b)	Define cut-set. Prove that every circuit in G has an even number of edges in common with any cut-set.	6	3	2
3. a)	Show that the complete tripartite graph $K_{1,2,3}$ is non planar.	5	3	2
b)	Write Fleury's algorithm supporting it with an example.	7	2	1

4. a) Find matching in the following graphs. Is it perfect matching? 6 2 1



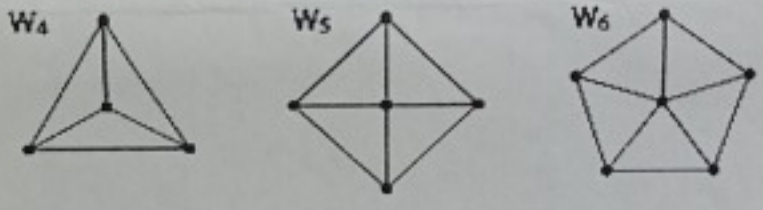
b) Prove that in a complete graph with n vertices there are (n-1)/2 edge disjoint Hamiltonian circuits, if n is an odd number and n >= 3 6 3 2

5. a) State Brook's theorem and Find vertex chromatic number for the following graph. 6 2 1



b) Explain Gupta-Vizing Theorem. 6

6. a) What is a planar graph? Show that the following graphs are self dual. 6 2 1



b) Show that a directed graph that contains an Euler circuit is strongly connected. 6 3 2

7. a) Prove that a connected planar graph with n vertices and e edges has e-n+2 regions. 6 2 2

b) Write Kruskal's algorithm? Find a minimal spanning tree in the following graph. 6 2 3

